Individual Pathways

Activity 10.3

Analysing energy doc-6095

1. **Types of energy and their sources**

   Complete the following table by identifying energy sources for each energy type listed.

   Sources: fluorescent tube, uranium fission, chocolate bar, moving tennis ball, vibrating violin string, rubber band, battery, infra-red lamp

Energy type	Source
Chemical	
Sound	
Light	
Electrical	
Heat	
Elastic potential	
Nuclear	
Kinetic	

2. **Types of potential energy**

   Categorise each of the types of potential energy shown in the table below.

Example	Category of potential energy
Air brakes in a truck	
Wound watch spring	
Kerosene	
Sugar	
Depressed trampoline canvas	
Comet	
Parachutist at the point of launch	

3. Objects of various masses (m) are raised from ground level to various heights (h) above the ground in a gravitational field. The following data were collected.

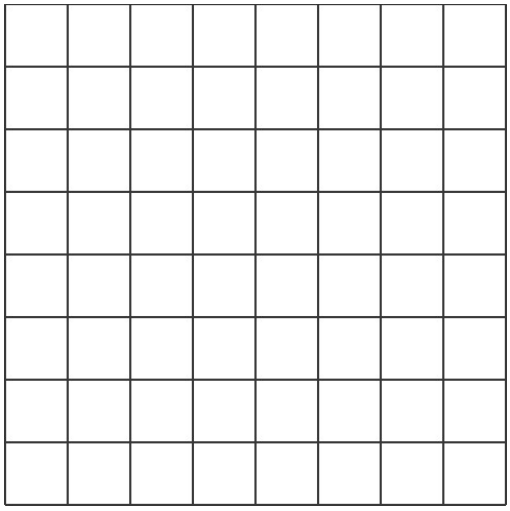
Mass: m (kg)	5	5	5	5	10	20	40
Gravitational acceleration: g (m/s^2)	9.8	9.8	9.8	9.8	9.8	9.8	9.8
Height: h (m)	2	4	6	8	4	2	1
Gravitational potential energy: GPE (J)							

   The formula for gravitational potential energy is: GPE = mgh.

   (a) Using the formula given above, calculate the GPE for each column.

© John Wiley & Sons Australia, Ltd

(b) Generate a scatter chart of GPE versus *h* in the space provided below. Complete the graph by inserting labels on each axis and an appropriate title.



(c) Outline the shape of this graph.

.....

(d) Describe how GPE changes for a fixed mass as it is raised to increasing heights above the ground.

.....

(e) Examine the data in the last four columns. Identify how the mass changes.

.....

.....

(f) Identify how the height changes.

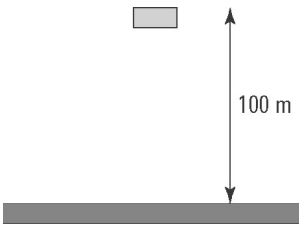
.....

.....

(g) Describe how these two changes affected GPE.

.....

4. An object of mass 10 kg is dropped from a stationary point 100 m above the ground, as shown below.



At any point during the fall, the object has gravitational potential energy (GPE) and kinetic energy (KE) given by the formulae:

© John Wiley & Sons Australia, Ltd

$$\text{GPE} = mgh \text{ and KE} = \frac{1}{2}mv^2$$

where m = mass (kg), g = 9.8 m/s^2, h = height (m) and v = speed (m/s).

(a) Use these formulae to complete the following table.

h **(m)**	**GPE (J)**	***v*** **(m/s)**	**KE (J)**	**GPE + KE (J)**
100	9800	0	0	9800
90	8820	14.000	980	
80	7840	19.799	1960	
70		24.249		
60		28.000		
50		31.305		
40		34.293		
30		37.041		
20			7840	
10		42.000		
0	0	44.272		

(b) Outline how GPE changes during the fall.

.....

(c) Outline how KE changes during the fall

.....

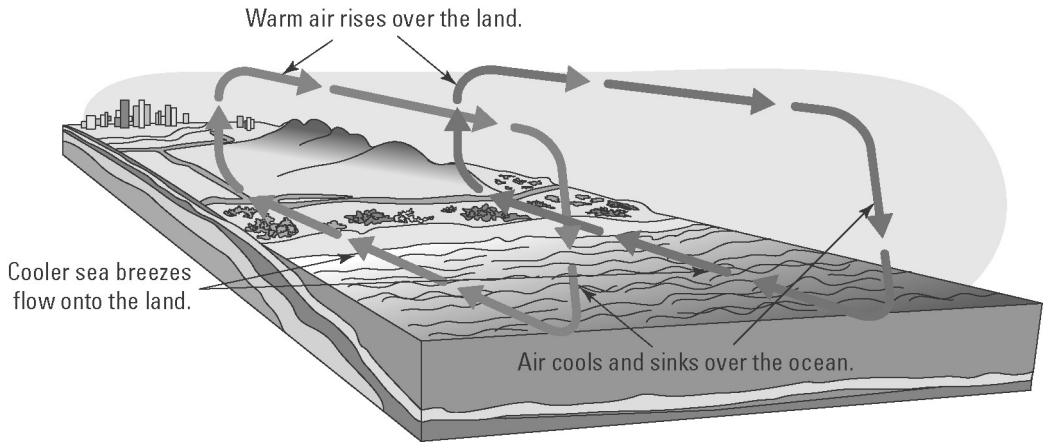
(d) Outline how the sum of GPE and KE changes during the fall.

.....

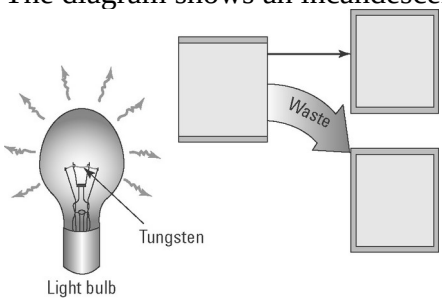
© John Wiley & Sons Australia, Ltd

5. **Convection**

Consider the following illustration showing the movement of air near the ocean shore. Describe how the production of convection currents makes the climate near the coast milder than it is inland.



6. The diagram shows an incandescent light bulb.



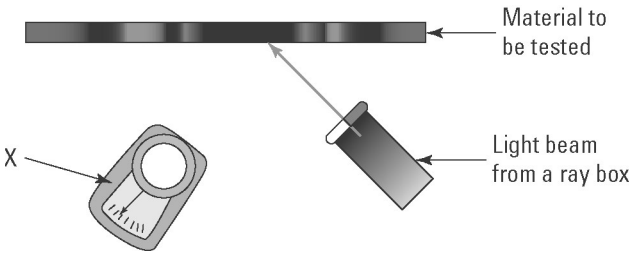
(a) Explain how an incandescent light bulb generates light.

(b) Insert the following words in the boxes in the diagram: heat energy, electrical energy, light energy

(c) Explain why incandescent light bulbs are not energy efficient.

7. The diagram below shows the procedure a student used to measure the intensity of reflected light from various materials and compare it with light reflected by a normal mirror.

© John Wiley & Sons Australia, Ltd



(a) Identify the measuring device X.

.....

(b) Why is a normal mirror also tested?

.....

.....

(c) Identify the variables that need to be controlled.

.....

.....

(d) The following materials were tested: shiny white cardboard, dull black cardboard and shiny black cardboard. Predict which cards will reflect the:

(i) most light energy

.....

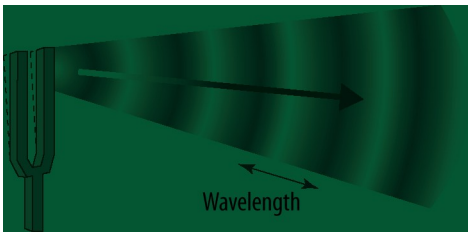
(ii) least light energy.

.....

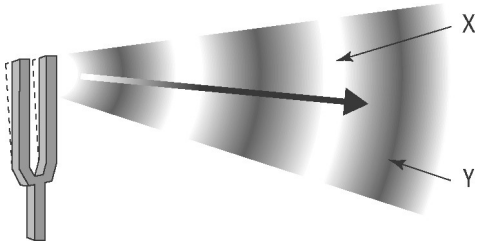
8. **Wavelength and frequency**

The following diagram shows two vibrating tuning forks producing sound waves.

A



B



(a) Identify the features of the wave labelled X and Y.

X:

Y:

(b) Which tuning fork produces sound waves with the lower pitch? Explain.

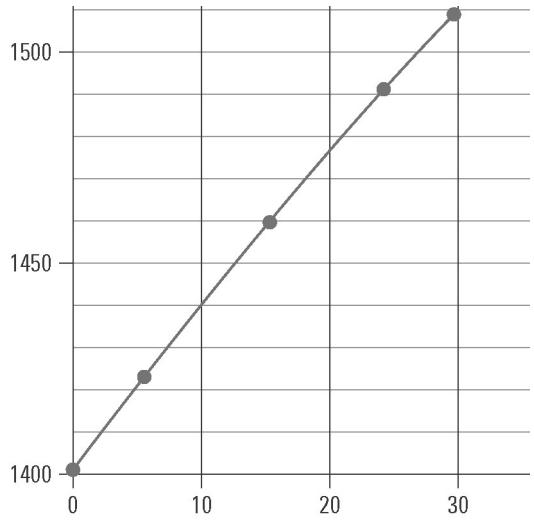
9. **Speed of sound in water**

© John Wiley & Sons Australia, Ltd

The following graph shows how the speed of sound (in m/s) in water varies with temperature (in °C).

(a) The labels and units on each axis are not shown. Insert these labels and units.

(b) Give the graph a title.



(c) Use the graph to determine the speed of sound in water at:

(i) 0 °C .....

(ii) 10 °C .....

(iii) 20 °C. .....

(d) In air, the speed of sound at 0 °C is 331 m/s.

(i) How many times faster does sound move in water than air at that temperature? (*Hint:* Divide one speed by the other.)

.....

.....

(ii) Explain why sound moves faster in water than in air.

.....

.....

.....

.....

.....

.....

.....

© John Wiley & Sons Australia, Ltd